



Enhancing Rural Transport Accessibility Through Public Transport Route Rationalization and Network Integration: Evidence from an Island-Province Case in the Philippines

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Abstract: The Public Utility Vehicle Modernization Program (PUVMP) is a key national reform in the Philippines’ mass transportation subsector. However, its application at the local level, island-provinces, has received limited attention. This study addresses that gap by evaluating Guimaras province’s Local Public Transport Route Plan (LPTRP). A questionnaire survey and transport modeling were used to assess travel behavior, accessibility, and network performance. Results show that many essential facilities, such as schools and health centers, are not adequately served by formal PUV routes. As a result, residents rely on informal modes that are often unsafe and expensive. The analysis also revealed issues with route overlap and inefficient area coverage. To address these local concerns, the study recommends redesigning routes, establishing transfer hubs, and adopting coordinated fleet management. These strategies aim to improve safety, accessibility, and system reliability for commuters. Overall, the findings offer a model for context-sensitive public transport planning in rural and island settings across the Philippines.

Keywords: Public transport planning; Rural accessibility; Route rationalization; Network integration; Transfer hubs; Fleet coordination

1 Introduction

The Public Utility Vehicle Modernization Program (PUVMP) is a nationwide mass transport reform initiative introduced by the Philippine government in 2017. The reform program aimed at comprehensively restructuring the country’s public transportation system. It positions both local and national government agencies as principal stakeholders in reorienting policies, institutional practices, and operational frameworks to ensure that public transport services meet standards of safety, reliability, environmental sustainability, affordability, and operational efficiency. Central to the program’s mandate is the resolution of longstanding structural challenges, including supply-demand imbalances and systemic operational inefficiencies, that have historically undermined the performance and quality of public transport services throughout the country.

Prior to the implementation of the PUVMP, the system allowed individual drivers and operators to function independently, often without coordination with other service providers along the same route. This lack of integration resulted in inconsistent service levels (i.e., oversupply during off-peak hours and severe undersupply during peak periods). Moreover, the competitive nature of on-street operations led to unsafe driving behavior, frequent unsanctioned stops, and a race for passengers, rather than the provision of quality service.

A key component of the PUVMP is the rationalization of routes through the development of Local Public Transport Route Plans (LPTRPs). Under the Omnibus Franchising Guidelines (OFG), franchise approvals now require a local government unit (LGU)-prepared LPTRP that reflects local transport needs [1]. In line with this, Guimaras Province developed its own LPTRP, outlining a proposed network of PUV routes to improve island-wide accessibility. These proposed routes, shown in Figure 1, are meant to provide structured and equitable public transportation services across the island.

According to the literature, islands are in a unique position when it comes to transportation because accessibility and the movement of people are critical to the balanced development of communities within islands [2–4]. Moreover, insularity extends beyond geographic isolation to include unique social and demographic traits, and within this context, transport is vital for development as connections depend on accessibility and communication, shaped by factors like scale economies and spatial reach of networks [2, 5, 6]. In essence, an efficient island transport network is crucial for economic, social, and spatial development, as the strong interdependence between inland and coastal areas requires integrated public and private transport planning that reflects local socio-demographic conditions and the physical characteristics of the territory [7–9]. Because of this, numerous studies have examined improvements in intra-island connectivity through enhanced public transport and Smart Island initiatives, showing that these lead to improved mobility and reduced transport inequalities [4, 10, 11].



Figure 1. Proposed transit lines in the Local Public Transport Route Plan (LPTRP) of Guimaras Province

However, while global studies have examined the role of intra-island connectivity, current studies on the implementation of the PUVMP have largely focused on urban settings or mainland provincial cities. Despite the Philippines being an archipelago, limited research exists on how the program is applied in insular contexts like Guimaras, an island-province in Western Visayas. While existing research has documented persistent challenges in LPTRP implementation across individual LGUs, these difficulties are in reality not isolated but reflect broader systemic issues in public transport reform [12–14]. Guimaras presents a particularly noteworthy case, not only due to its insular geography and dispersed population, but because its LPTRP represents a consolidation of multiple LGU-level plans into a single island-wide framework. This scale of implementation offers a distinct lens through which the complex challenges of route rationalization, demand planning, and service coverage can be examined alongside the unique constraints that arise when coordinating transport reform across an entire island-province.

Thus, this study aims to review the local government’s proposed PUV routes in Guimaras Province, identify service gaps, and highlight potential areas for improvement. As an exploratory study, the primary objective is not to conduct a detailed operational analysis, but rather to offer an assessment of the current LPTRP proposal and provide a general framework that can serve as a baseline for program implementation at the provincial level. Through transport modeling and accessibility analysis, the research seeks to provide planning guidelines and actionable insights for designing inclusive and context-sensitive public transport networks in rural and island settings. The findings are intended to be applicable beyond Guimaras, offering considerations relevant to other island-provinces across the

Philippines and similar insular settings in Southeast Asia where comparable transport reform programs are being pursued. Ultimately, this study seeks to contribute to a more equitable and sustainable mobility system that meets the needs of all residents, regardless of location.

2 Review of Related Literature

2.1 Implementation of Public Utility Vehicle Modernization Program

The PUVMP represents a significant and ambitious reform initiative within the Philippine transport sector, aiming for a wholesale transformation of the public land transportation industry. The program was primarily conceived to address long-standing and complex issues such as severe traffic congestion, high carbon emissions, and a high incidence of accidents, all of which are persistent problems in urban mobility globally. Beyond environmental and economic concerns, social pressures related to commuter welfare and the general acceptance that existing transport systems needed to change also fueled the program's creation. The PUVMP mandates the modernization of vehicles to meet Euro IV emission standards or better, the inclusion of modern features like GPS, automatic fare collection systems, CCTV cameras, and a shift towards consolidated transport operations [1].

The provision of public transport services in the Philippines has largely depended on private sector initiative, wherein operators identify a business opportunity, apply for a franchise, and provide service along a particular route with little to no system-level planning involved. The absence of integrated planning ultimately leads to sub-optimal network coverage, unreliable service, and broader externalities such as congestion and environmental degradation. The PUVMP seeks to address this by shifting the lead in transport planning and service provision to local governments. Under this framework, LGUs are required to conduct route rationalization studies and prepare an LPTRP, which serves as a prerequisite for franchise approval within their jurisdiction, ensuring that service provision is grounded in actual passenger demand and travel patterns [15].

Despite its intended benefits, the program has been subject to sustained negative scrutiny from jeepney operators, commuters, and the broader public. According to Mateo-Babiano et al. [16], several transition-related challenges were identified by operators, including concerns regarding the government's capacity to effectively manage and oversee sectoral reforms. Smaller operators tend to exhibit resistance to change and express apprehension about the risks posed to their existing business models, often expecting greater institutional support to mitigate these uncertainties. Additionally, there is a recognized need for capacity-building initiatives, as many operators lack the necessary skills to adapt to the proposed contracting framework. From the commuters' perspective, assessments of the PUVMP suggest that satisfaction levels are generally limited to the fulfillment of basic service expectations rather than the delivery of enhanced transport experiences [17]. Persistent issues remain evident, including service inefficiencies related to vehicle availability, overcrowding, hygiene, and route suitability, indicating that significant gaps in service quality continue to affect daily users despite modernization efforts. More broadly, the program has also prompted public dissent, particularly among jeepney operators and drivers, culminating in organized transport strikes that directly affected commuters. Public discourse surrounding these events reflects a complex response, with social media sentiment showing notable support for the grievances raised by transport groups while also expressing opposition to the jeepney phase-out scenario associated with the modernization program [18].

In terms of actual operational services, the rollout of the PUVMP in Metro Manila has been especially critical given the region's severe transport issues, including high congestion costs and worsening air pollution [1]. Despite the introduction of rationalized routes, overlapping services and operational inefficiencies continue to persist, reflecting the difficulty of restructuring an already dense and highly competitive transport network. Similar issues are evident in provincial areas, where studies on LPTRP implementation point to several structural and operational constraints [12–14]. These include financial pressures on operators due to restricted or restructured routes, outdated demand assumptions embedded in initial LPTRP formulations, and the high capital costs associated with fleet modernization. In addition, informal transport services continue to play a significant role, often filling gaps left by formal plans but also undermining regulatory objectives. The persistence of underutilized and overlapping routes further indicates that, despite the shift toward planned provision, coordination challenges remain unresolved.

While existing research has largely concentrated on LPTRP implementation in individual LGUs, the persistent challenges documented across these cases suggest that difficulties in execution are not isolated but systemic. In this regard, Guimaras presents a particularly noteworthy case, not only due to its island geography and dispersed population, but because its LPTRP represents a consolidation of multiple LGU-level plans into a single island-wide framework. This scale of implementation offers a distinct lens through which the compounded challenges of route rationalization, demand planning, and operator compliance can be examined, alongside the unique opportunities and constraints that arise when coordinating transport reform across an entire island-province.

2.2 Rural Accessibility

Rural accessibility is a key concern in global development agendas, particularly under the Sustainable Development Goals (SDGs). SDG Target 9.1 emphasizes the provision of quality, reliability, sustainability, and resilient

infrastructure that ensures affordable and equitable access for all. Its associated indicator, SDG 9.1.1, measures the proportion of the rural population living within two kilometers of an all-season road, highlighting basic physical connectivity as a prerequisite for economic and social participation. Complementing this, SDG 11.2.1 measures the proportion of the population with convenient access to public transport, shifting the focus from the mere presence of roads to the availability and convenience of actual transit services [19]. While SDG 9.1.1 is concerned with whether rural communities have a physical road connection to broader networks, SDG 11.2.1 addresses whether people can readily access public transportation once that physical connection exists. Together, these indicators reflect a progression of accessibility. This involves both providing roads that connect rural communities to broader networks and ensuring the availability of reliable public transport within reasonable proximity. Although convenient transit access is more commonly associated with urban areas, the provision of equitable public transportation remains a fundamental goal regardless of geographic context. Although the latter is more commonly associated with urban contexts, equitable access to public transportation remains a broader goal regardless of geographic setting. In terms of what constitutes reasonable access, studies generally indicate that the acceptable walking distance to a transit stop ranges from 400 m to 800 m, with 500 m commonly cited as a practical minimum or average threshold across varying contexts [20–23]. This pattern extends to rural settings as well, where travel preferences align with distance, with walking being the preferred mode for short trips of up to 500 m [24].

However, accessibility extends beyond physical infrastructure. As noted by Brovarone and Cotella [25], it also involves reducing the need for travel by bringing services closer to communities, strengthening local capacities, and supporting localized development. Transport infrastructure nonetheless remains a major driver of growth in low- and middle-income countries, influencing agricultural productivity, market access, livelihoods, and access to healthcare and education, while also shaping environmental outcomes. Kaiser and Barstow [26] show that improved rural transport supports economic development, agriculture, health, education, gender equality, and environmental sustainability. These benefits can also extend beyond local areas, as infrastructure investments reduce food insecurity in both host and neighboring districts by improving agricultural flows and market access [27]. In healthcare, systemic transport barriers can intensify existing inequalities. Neely and Ponshunmugam [28] show that in South Africa, transport networks favor commercial activity over healthcare access in rural areas, forcing residents to rely on costly private transport or endure ambulance delays, thereby deepening health disparities. While kinship networks sometimes mitigate these barriers, they also reflect broader structural issues.

In the Philippine context, rural transport infrastructure development, such as roads and bridges, directly benefits rural farmers and local economies. Road infrastructure plays a critical role in connecting rural communities to markets, reducing input and transaction costs, and supporting agricultural productivity. In fishing communities, improved road access has similarly been shown to generate considerable benefits, enabling better market connectivity and supporting increases in production and productivity [29]. Llanto [30] further affirms that regions with greater infrastructure investment tend to experience higher economic growth, while those with inadequate road networks remain at a disadvantage in terms of productivity and competitiveness. Beyond economic activity, transport infrastructure also has direct implications for access to social services, particularly healthcare. In island and interior communities, poor road conditions and geographic isolation have been associated with travel times exceeding 60 minutes to the nearest health facility, placing these populations at a significant disadvantage in terms of health equity [31, 32]. Where road access is limited, communities are not only physically distant from hospitals and rural health units, but are effectively excluded from timely medical attention, which disproportionately affects vulnerable groups such as women, children, and the elderly. Improving road connectivity in these areas, therefore, serves a dual function, by both facilitating economic activity and enabling communities to reach essential health services within acceptable travel times. Studies examining rural road conditions further highlight the need for continued infrastructure improvements to support safe and efficient movement in these areas [33]. Ultimately, the enhancement, improvement, and extension of rural transportation infrastructure bring significant and multifaceted benefits to rural communities, improving their quality of life and fostering sustainable development [26].

2.3 Improving Rural Route Accessibility

Accessibility itself is a multidimensional concept, encompassing the physical transport system in terms of travel time and cost, the spatial distribution of activities, the temporal availability of services, and the individual characteristics of users, including their socioeconomic and demographic circumstances [34]. Taken together, these dimensions highlight that proximity to a road alone does not guarantee meaningful access, and that equitable accessibility requires transport systems that are responsive to the actual needs and constraints of the populations they serve. As mobility planning shifts away from car-centered models, the limitations of rural transport systems become more evident. Low population densities often lead to infrequent public transport services, long waiting times, and extended travel durations, forcing many households to rely on private vehicles despite unequal access. Residents of rural areas travel to work, schools, health facilities, and markets just as frequently as their urban counterparts, yet they cannot benefit from the transport infrastructure and service density available in cities, leaving informal transport

as the predominant option due to its lower cost in less populated areas [35]. This highlights the need to strengthen public transport as a core component of equitable rural accessibility [36].

Rural areas also face distinct accessibility challenges compared to urban settings due to geography and settlement patterns. In Chungju-si, Korea, Jeon et al. [37] found that nearly 20% of villages were more than 10 minutes' walk from the nearest bus stop, with low bus speeds and infrequent services limiting access to essential facilities. Emergency centers had the longest access times, followed by health centers and schools, with mountainous and low-density areas experiencing the poorest accessibility. This emphasizes how geography, infrastructure, and service frequency together drive transport inequalities in rural areas, and that improving accessibility requires addressing each of these dimensions rather than physical distance alone.

Brovarone and Cotella [25] argue that improving rural accessibility requires a multilayered policy approach that goes beyond physical distance. They emphasize that conventional public transport in rural areas is often unsustainable and subject to service cuts, reinforcing social exclusion. Key challenges include socioeconomic disparities that lead to car dependency and exclude vulnerable populations such as the elderly and low-income households, institutional resistance to flexible service models, and limited digital infrastructure that constrains the rollout of smart mobility solutions. It is also worth noting that greater accessibility, while necessary, is not in itself a sufficient condition for social justice, as the economic costs and environmental impacts of providing that accessibility must also be considered in determining what constitutes a viable and equitable solution [38]. Similarly, Truden et al. [39] argue that good reachability for rural populations is not solely defined by walking distance to the nearest stop but also by geographical factors such as elevation, and that routes should be designed to be accessible through integrated and connected transport options that account for the specific geographical and social contexts of rural areas.

One approach that has gained attention in rural transport planning is the use of multimodal mobility hubs, which link new mobility services with existing public transport at key nodes, enabling more convenient intermodal transfers and expanding the range of feasible travel itineraries. Beyond their functional role, these hubs can also provide supporting infrastructure such as shelters, signage, and waiting facilities that improve the overall passenger experience. Decisions regarding their placement should be guided by their ability to improve access to key destinations including workplaces, health facilities, markets, and schools, rather than simply reflecting existing traffic volumes [36]. Demand-responsive transport has also been proposed as a complementary solution for areas where fixed-route services are not operationally sustainable, offering flexible scheduling that can adapt to low and variable demand. However, its implementation has not been without difficulty, and many demand response transit (DRT) initiatives have failed in practice due to the complexity of aligning flexible service models with varying local conditions [40]. Its outcomes vary considerably depending on the socioeconomic, cultural, and institutional context of the area [25], and evidence suggests that in some rural settings, a well-planned conventional network with fixed scheduled routes may ultimately offer more simplicity, reliability, and convenience than demand-responsive alternatives [41].

In the Philippines, the informal nature of public transport in remote areas, further complicates accessibility. A common mode of transport is the *habal-habal*, which refers to a motorcycle modified to carry multiple passengers along rural roads. This mode of transport, while ubiquitous and often the only option, highlights the lack of formal, regulated public transport services in many rural barangays. This reliance on informal transport directly impacts the daily lives of professionals working in these remote locations. For instance, Flores et al. [42] examine factors affecting the retention of healthcare workers in the Philippine National Rural Physician Deployment Program (Doctors to the Barrios). While their study covers multiple factors, it implicitly points to the significant travel burden faced by these doctors, who often have to commute long distances from more accessible areas to reach their assigned rural barangays. Similarly, Equipado and Asis-Gilbas [43], in their study on the experiences of elementary teachers in a remote school in Sorsogon, discuss transportation as a key challenge affecting teachers' daily sustenance and their ability to deliver instruction. These studies collectively illustrate how the absence of robust, formal public transport networks in rural Philippines necessitates long and often difficult journeys for essential service providers, impacting their work and well-being.

3 Methodology

3.1 Study Area

Guimaras, officially designated as the Province of Guimaras, is an island-province situated within the Western Visayas region of the Philippines. The province is administratively composed of five municipalities, namely Buenavista, Jordan, Nueva Valencia, San Lorenzo, and Sibunag, and is further subdivided into 98 barangays. Jordan serves as the provincial capital, while Buenavista is the most extensive municipality in terms of both land area and population. Geographically, the province occupies a position within the Panay Gulf, between the islands of Panay and Negros, bordered to the northwest by the province and city of Iloilo and to the southeast by Negros Occidental. Guimaras forms part of the Metro Iloilo–Guimaras area, one of twelve officially recognized metropolitan regions in the country.

The provincial territory encompasses Guimaras Island as its primary landmass, alongside several smaller surrounding islets, including Inampulugan, Guiwanon, Panobolon, Natunga, and Nadulao. Despite its relatively compact geographic extent, the province exhibits a dispersed settlement pattern characterized by inland, coastal, and upland communities. The local economy is predominantly driven by agriculture, fisheries, and tourism, underpinned by a network of secondary and circumferential road infrastructure. Nevertheless, transport connectivity remains constrained, with a considerable number of barangays dependent on a limited number of primary road corridors, most notably the Guimaras Circumferential Road, for inter-barangay and inter-municipal mobility.

A critical consideration in the formulation of the revised route plans was the spatial distribution of essential socio-economic facilities and activity generators across the province. An inventory of key establishments, encompassing schools, health centers, public markets, government offices, and other public service facilities, was conducted to identify major activity nodes within the network. The precise locations of these facilities were obtained and subsequently mapped within a geospatial environment. Figure 2 illustrates the geographic extent of Guimaras Province and the spatial distribution of these essential facilities across the island.

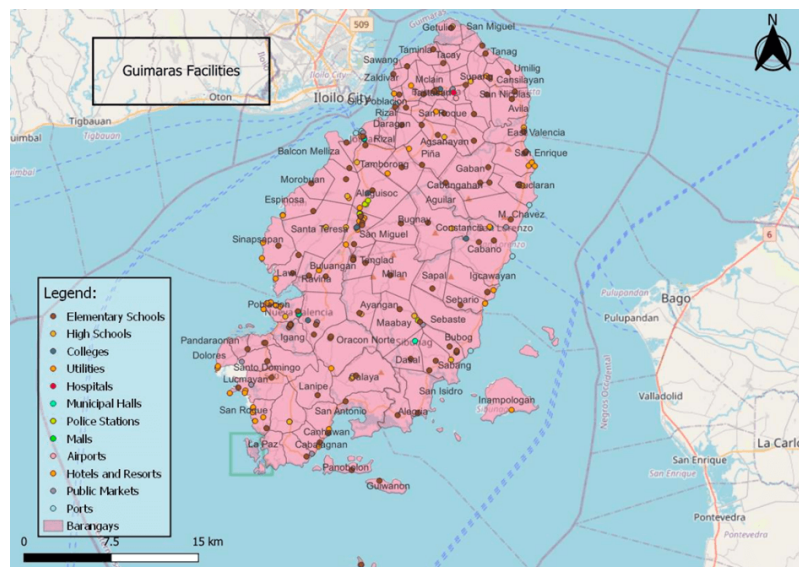


Figure 2. Guimaras Island facilities map

3.2 Questionnaire Survey

A Home Interview Survey (HIS) was conducted in 2018 to systematically document the day-to-day travel behavior of residents across Guimaras Province. The primary objective was to establish a comprehensive understanding of commuter movement patterns and to provide an empirical foundation for the analysis of local mobility dynamics. Data collection was carried out across multiple locations throughout the province and at varying times of day to ensure that the resulting dataset adequately represented the breadth of travel activities occurring within the study area. The sampling framework employed a stratified approach, wherein households were selected from multiple barangays spanning all five municipalities, ensuring representation of both densely populated urban centers and more sparsely settled rural communities. While the survey instrument was developed in English, enumerators were trained to administer the questionnaire in the local dialect to facilitate comprehension and improve response accuracy, with responses subsequently encoded into the survey forms.

The questionnaire was structured into several thematic sections encompassing socio-demographic characteristics, household composition, travel behavior, and transport-related expenditures. Socio-demographic variables collected included age, sex, income level, household size, and vehicle ownership. Travel-related items covered trip purpose, travel time, departure and arrival times, primary and secondary modes of transport, access and egress times, and out-of-pocket trip costs. Respondents were further asked to identify their origin and destination barangays, enabling the spatial mapping of trip flows within the province. Additional survey items gathered data on the access modes used to reach public utility vehicle stops or terminals, as well as respondents' ranked preferences for public transport service attributes and amenities.

Upon completion of data collection, responses were encoded, cleaned, and validated using Microsoft Excel, then anonymized and aggregated prior to analysis. The processed survey data served as the primary basis for characterizing the socio-demographic profile of the resident population of Guimaras Province and for delineating prevailing travel patterns across the region.

3.3 Transport Model

Using Equilibre Multi-Modal/Multi-Modal Equilibrium (EMME) transport planning software, a network model of Guimarães was developed that included the full extent of the province's road system. This model consisted of nodes and links representing intersections and road segments. Zoning boundaries were defined based on barangay and municipal divisions, grouping areas according to geographic proximity and connectivity. Available transportation modes within the province were also encoded into the model to accurately reflect existing mobility conditions. In addition, the proposed transit lines from the Guimarães Local Government, as shown earlier in Figure 1, were individually inputted and modeled to represent their routing and intended operational characteristics.

To ensure the model accurately represented the operational characteristics of the study area, each road type was assigned specific parameters, including free-flow speed, lane capacity, and a volume–delay function (VDF). These model parameters are summarized in Table 1. The 20 km/h speed assigned to zone connectors reflects the prevailing conditions of the access links in the study area, which consist of local roads, earth roads, and roadside friction-heavy corridors. These connectors are generally underdeveloped and serve primarily as network completion links rather than functional travel corridors. In practice, only auxiliary and informal transport modes utilize these links, including walking, cycling, and informal paratransit such as tricycles and habal-habal. The 20 km/h value therefore represents a composite speed across all access modes considered in the study.

Table 1. Model parameters

Type	Length (km)	Free-Flow Speed	Lane Capacity (veh/hr/lane)
Primary national road	15.64	60	900
Tertiary national road	285.55	40	800
Collector road	88.36	40	700
Local/service road	327.11	30	500
Zone connector	-	20	9999

The VDF is particularly important, as it captures how road performance varies according to functional classification. Major roads were assigned more efficient VDFs to reflect their higher capacities and smoother traffic flow, whereas collector and local roads were given less efficient VDFs to account for roadside friction, narrower carriageways, and frequent interruptions. In this study, the volume–delay relationship follows the Bureau of Public Roads (BPR) function, expressed as:

$$T = T_f \left[1 + \alpha \left(\frac{q}{C} \right)^\beta \right] \quad (1)$$

where, T is the actual travel time on a road, while T_f is the free-flow travel time. The value q represents the actual traffic volume, and C represents the road's capacity. The two parameters, α and β , determine how quickly travel times increase as a road gets closer to its maximum capacity. In practice, this means that when the road is lightly used, travel times stay close to free-flow levels, but as traffic builds up, travel times increase sharply. Table 2 provides the VDFs used in this study, expressed in the format understood by the software.

Table 2. Volume–delay functions (VDFs) used per road classifications

Type	Travel Time Expression (minutes)
Primary National Road	$\frac{L \cdot 60}{e_1} \left[1 + 0.6846 \left(\frac{V_{au} + V_{ad}}{n_{lanes} e_2} \right)^{3.1644} \right]$
Tertiary National Road	$\frac{L \times 60}{e_1} \left[1 + 0.7312 \left(\frac{V_{au} + V_{ad}}{n_{lanes} \times e_2} \right)^{3.6596} \right]$
Collector Road	$\frac{L \cdot 60}{e_1} \left[1 + 0.7312 \left(\frac{V_{au} + V_{ad}}{n_{lanes} e_2} \right)^{3.6596} \right]$
Local/Service Road	$\frac{L \cdot 60}{e_1} \left[1 + 0.8774 \left(\frac{V_{au} + V_{ad}}{n_{lanes} e_2} \right)^{4.4613} \right]$
Zone Connector	$\frac{L}{e_1/60}$

In this formulation:

- L —the length of the road segment (km);
- $V_{au} + V_{ad}$, (volau+volad)—total volume of private (volau) and public (volad) vehicles on the road segment (veh/hr);
- e_1 —design free-flow speed (kph);
- e_2 —roadway capacity per lane (veh/hr/lane); and
- n_{lanes} —number of lanes in the segment.

To provide a typical comparison about the shape of the VDFs between the primary national road (Figure 3a) and the local/service road (Figure 3b) where the vertical axis represents the travel time (in minutes) along the segment given the volume (veh/hr) along the x -axis, using the former consumes less time than the latter when vehicular volume reaches around 1000 veh/hr.

Trip distribution patterns were derived using an Origin–Destination (OD) matrix developed from the household questionnaire survey. The OD pairs were defined at the barangay level, which represents the smallest administrative unit in the Philippines, comparable to neighborhoods or districts, with a total of 100 barangays considered in the study area. This matrix was subsequently loaded into the EMME model to generate an OD histogram. Figure 4 illustrates the resulting OD matrix for jeepney users. The results indicate that both trip origins and destinations are primarily concentrated within the main urban areas of Guimaras and around key port locations connecting to Iloilo City. This pattern reflects the tendency of jeepney demand to cluster in densely populated and economically active areas.

To enhance model accuracy, inputs were calibrated using actual transport metrics, including observed travel times, trip volumes, and the availability of vehicle units. Key attributes such as vehicle capacity and average service headways for each mode were based on the LPTRP Manual Volume 1. The Transit Demand Adjustment (TDA) tool was employed to iteratively refine transit demand using a gradient-based method. Starting from the initial OD matrix and observed volumes on selected transit segments, links, or nodes, the TDA adjusts the matrix to better match actual demand. After each adjustment, the simulated link volumes are then compared to observed counts. Systematic differences are expressed as scaling multipliers applied to the adjusted OD matrix. This cycle of adjustment, assignment, comparison, and scaling continues until the statistical fit, measured by the coefficient of determination (R^2), meets or exceeds a predefined threshold. A simplified methodological framework of this process is presented in Figure 5.

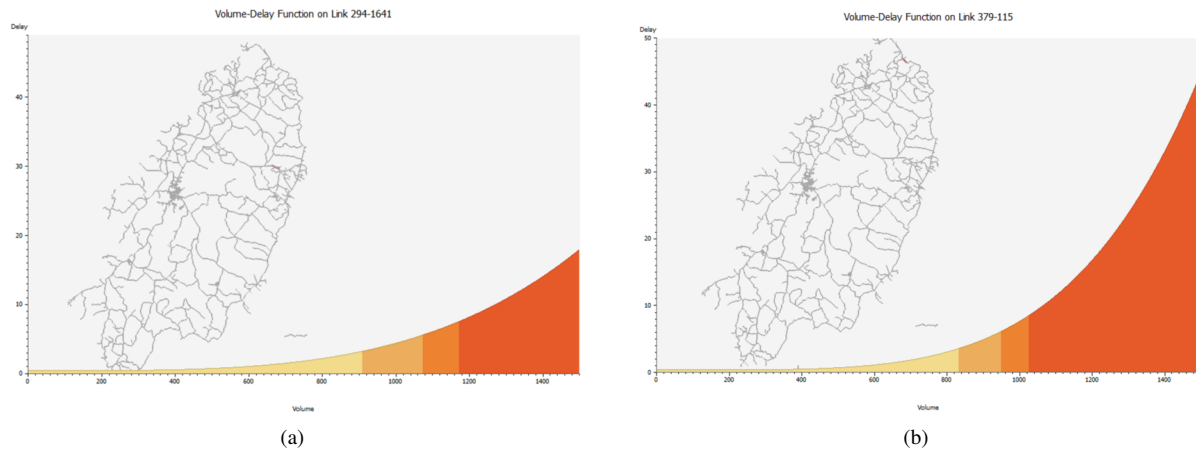


Figure 3. (a) Example of a volume–delay function (VDF) along a primary national road; (b) example of a VDF along a local road

Passenger counts were conducted at the origin terminal of each jeepney route, capturing boarding volumes at the initial departure point of every line. No data were collected at intermediate stops along the routes, which constitutes a limitation of the study. As passengers may board and alight at multiple locations, the absence of mid-route observations means that localized variations in demand along each corridor may not be fully represented in the model. Despite this, focusing on origin terminals provided a practical and consistent basis for data collection, as these locations typically exhibit the highest passenger concentrations. In the Philippine context, jeepneys commonly depart only after reaching near-full capacity at their origin terminals, making these points a reasonable proxy for total route-level demand. The calibration results are presented in Figure 6, where observed hourly passenger volumes (y -axis) are compared against model-predicted volumes (x -axis). The data points generally lie close to the line of perfect agreement, indicating a satisfactory level of model fit. This is supported by R^2 of 0.8628 obtained from linear regression, suggesting that approximately 86.28% of the variation in observed passenger volumes is explained

by the model. It should be noted that there was an outlier located near one of the port areas. This is likely due to the highly variable demand in the vicinity, as vessel arrivals and departures cause irregular surges and lulls in jeepney ridership that the model does not fully capture.

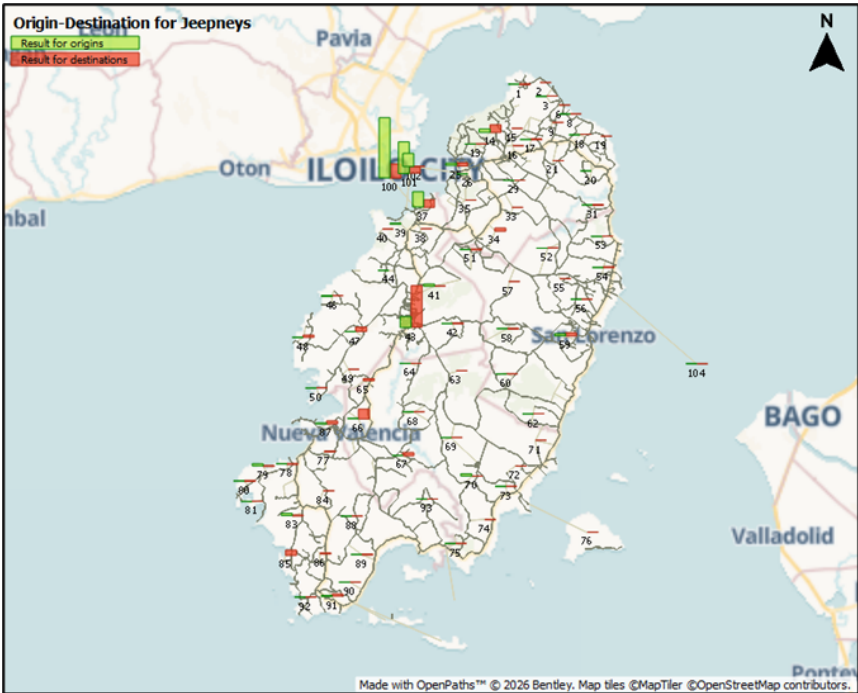


Figure 4. Spatial distribution of Origin–Destination (OD)

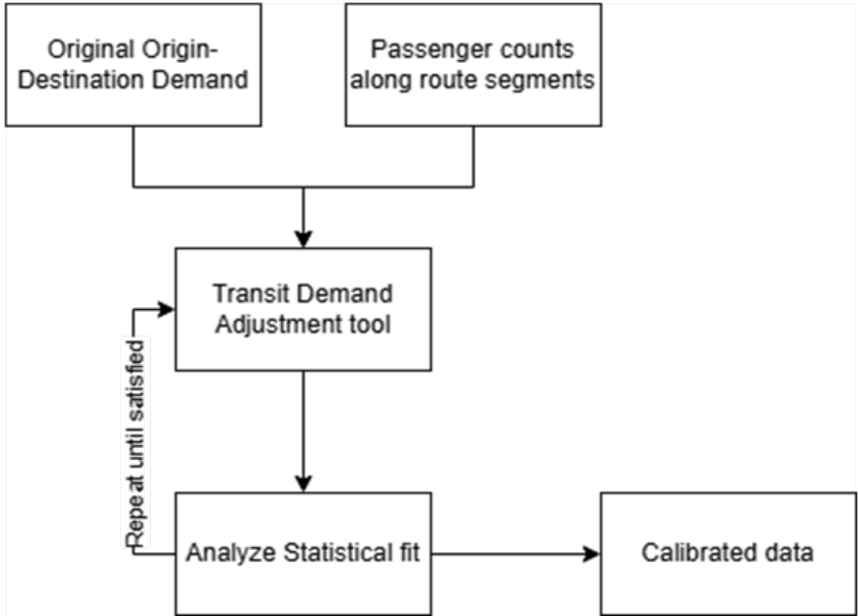


Figure 5. Flowchart for transit demand adjustment

Following calibration of the 2018 baseline model, demand for 2025 was projected using vehicle-specific growth rates from the Project Preparation Division-Planning Services, Department of Public Works and Highways (DPWH) (see Table 3), with jeepneys assigned their corresponding rate. These data were used to compute a weighted growth rate by multiplying each mode’s peak hour volume by its respective growth rate and deriving the overall average. To estimate the growth rate for the study year 2026, linear forecasting was applied based on the available data. Using this approach, the projected annual traffic growth rate from the 2018 baseline to 2026 was determined to be 7.32%.

However, it should be noted that the use of the 2018 HIS dataset was necessitated by the absence of more recent

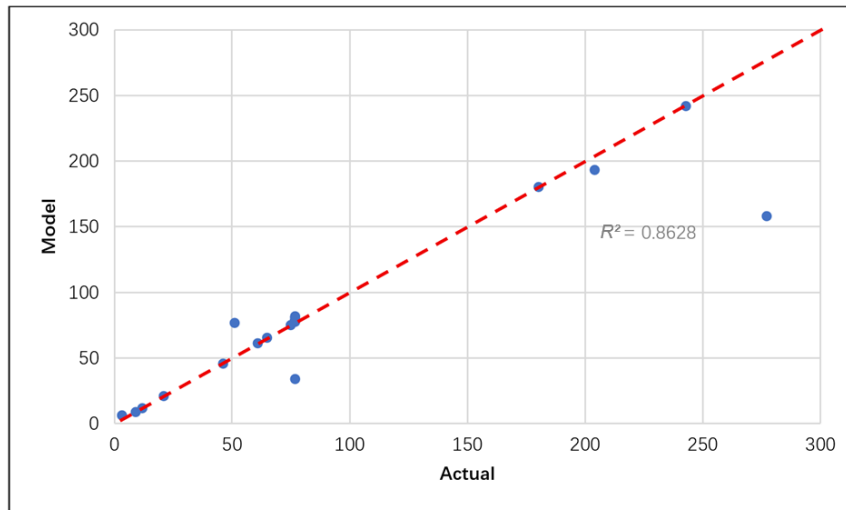


Figure 6. Calibration of the network model

Table 3. Department of Public Works and Highways (DPWH) data on projected growth rates

Year	Cars/Pickup/Vans	Jeepneys/Buses	Trucks	Motor/Tricycle
2015–2020	9.24%	8.03%	6.00%	6.41%
2020–2025	9.19%	7.97%	5.94%	6.35%
2025–2030	8.98%	7.77%	5.75%	6.15%
2030–2035	8.75%	7.54%	5.52%	5.92%
2035–2040	8.54%	7.33%	5.32%	5.72%
2040–2045	8.35%	7.14%	5.13%	5.53%

island-wide origin-destination survey data. While demand growth adjustments were applied to account for temporal changes in travel demand, these projections primarily reflected aggregate traffic growth trends and may not have fully captured post-pandemic behavioral shifts in mobility patterns. Such changes may include the effects of remote work arrangements, online education, tourism recovery dynamics, and evolving modal preferences following the COVID-19 pandemic. Consequently, the projected travel demand estimates may not completely represent current travel behavior across the province. Future studies and subsequent LPTRP updates may benefit from the conduct of updated household travel surveys or the integration of mobile-based mobility datasets to better capture evolving travel behavior patterns, particularly within insular and rural contexts.

Rural accessibility, in this context, refers to the degree of ease with which residents can reach essential services such as schools, health centers, commercial areas, and government offices through the available transport network. Using the provincial transport network database, a shortest distance analysis was performed to determine the average, nearest, and farthest travel distances from each barangay to the nearest facility of each type. These metrics were derived from the calibrated network model and served as indicators of spatial accessibility for each municipality. The results provided a quantitative basis for assessing transport equity across the island. In this manner, the transport model translated the abstract concept of rural accessibility into measurable indicators that could be analyzed under varying operational policies.

4 Results and Discussion

4.1 Demographics

The socio-demographic profile of respondents (see Table 4) suggests that the survey largely captures the perspectives of young to middle-aged adults. This indicates that the findings are strongly reflective of the population segment most engaged in daily commuting for work, education, and other routine activities. As such, the results are likely to have an influence on more practical concerns such as travel time, cost, reliability, and accessibility, which are typically more salient for regular transport users. The gender distribution shows a slightly higher representation of female respondents. Women often exhibit different travel patterns and sensitivities compared to men, including greater reliance on public transport and heightened concern for personal security. Household size trends indicate that most respondents belong to medium-sized households, suggesting shared economic responsibilities and potentially constrained household budgets. This aligns with the income distribution, where a large proportion falls within lower-

to middle-income brackets. Together, these characteristics point to a user base that is likely sensitive to transport costs and dependent on affordable mobility options. In this context, public transport is a necessity, which reinforces the importance of maintaining equitable transportation systems. At the same time, the concentration of respondents within modest income ranges highlights potential challenges in the adoption of modernization policies that may increase travel costs, such as fleet upgrades. Financial constraints among users may limit their willingness or ability to absorb fare increases, while also reflecting similar constraints among operators who serve these markets. Overall, the demographic profile suggests that the survey captures a population that is highly reliant on public transport, cost-sensitive, and engaged in routine, necessity-driven travel.

Table 4. Socio-demographic data of respondents ($N = 2195$)

Variable	Category	Data
Age (year)	1–10	4 (0.18%)
	11–20	108 (4.92%)
	21–30	862 (39.27%)
	31–40	520 (23.69%)
	41–50	337 (15.35%)
	51–60	219 (9.98%)
	61–70	114 (5.20%)
	71–80	23 (1.05%)
	≥81	8 (0.36%)
	Mean	35.72
Gender	Male	884 (40.27%)
	Female	1284 (58.50%)
	Others	27 (1.23%)
Household size (number of persons)	1–3	475 (21.64%)
	4–6	1474 (67.15%)
	7–9	214 (9.75%)
	10+	32 (1.46%)
	Mean	4.72
Monthly personal income (Peso/month)	<2500	408 (18.68%)
	2501–5000	549 (25.14%)
	5001–7500	698 (31.96%)
	7501–10000	235 (10.76%)
	10001–15000	163 (7.46%)
	15001–20000	56 (2.56%)
	20001–25000	45 (2.06%)
	25001–30000	20 (0.92%)
	30001–40000	4 (0.18%)
	40001–50000	1 (0.05%)
	50001–75000	2 (0.09%)
	75001–100000	1 (0.05%)
	100001–150000	1 (0.05%)
	≥150001	1 (0.05%)
	Mean	6531.59

On the other hand, Table 5 presents the data on transport mode and trip purpose among respondents. It could be observed that the population has a strong reliance on informal and semi-formal mobility options, particularly motorcycles and tricycles. This suggests that the local transport environment is highly flexible and responsive to dispersed travel demand but also fragmented. The prominence of motorcycles, in particular, points to a preference for speed, convenience, and point-to-point travel, especially in areas where formal public transport services may be limited or inefficient. At the same time, the continued use of jeepneys highlights their enduring role as an accessible and affordable mode, even amid ongoing modernization efforts. The relatively low share of higher-capacity modes such as buses and UV/FX services implies that mass transit options are either limited in availability or not well aligned with users' travel needs. This reinforces the notion that the existing system is dominated by smaller, more adaptable vehicles rather than coordinated, high-capacity networks. Walking, although representing a smaller share, still reflects the importance of short-distance trips and the role of proximity in daily mobility. This suggests that the spatial distribution of services influences travel behavior, with some trips remaining within walkable distances.

Table 5. Transport mode and trip purpose of respondents

Transport Mode	Frequency (%)	Trip Purpose	Frequency (%)
Own passenger car	78 (3.55%)	Work	1252 (57.27%)
Own motorcycle	859 (39.13%)	School	151 (6.91%)
Tricycle	519 (23.64%)	Market/Shopping	693 (31.70%)
Trisikad	45 (2.05%)	Recreation	28 (1.28%)
UV/FX/Shuttle	8 (0.36%)	Church/Mosque	13 (0.60%)
Bus	15 (0.68%)	Others	49 (2.24%)
Jeepney	474 (21.59%)		
Habal-habal	66 (3.01%)		
Multicab	2 (0.09%)		
Others	38 (1.73%)		
Walking	91 (4.15%)		

The relationship between income and transport mode choice in Guimaras Province reflects the limited fiscal capacity of most residents. With a mean monthly personal income of around P6,500, most respondents belong to low-income households that cannot afford high-cost modes such as private cars. This economic limitation is evident in the dominance of motorcycles, which serve as the most common private vehicle due to their low purchase and operating costs. In contrast, jeepneys function as the primary formal public transport option, offering fixed routes that cater to daily commuting needs. Meanwhile, tricycles and habal-habal are classified as informal public transport modes, providing short-distance services where formal routes are unavailable. The high usage of these modes indicates that affordability and accessibility take precedence over comfort or regulation in travel decisions. This highlights that income is a strong determinant of mode preference. Lower-income individuals depend heavily on affordable informal and public transport, while those with slightly higher earnings opt for motorcycles as a low-cost private alternative. The minimal share of private car users underscores the need to strengthen affordable, reliable, and safe public transport options across the province.

4.2 Route Accessibility

Figures 7a and 7b illustrate the overlain map of essential facilities in Guimaras Province in relation to the proposed PUV routes. The yellow lines represent the proposed transit corridors, while the blue shaded areas indicate the catchment zones of each transit line within a 500-meter radius, signifying the areas with direct access to public transport services. This threshold is consistent with established literature, which generally identifies 500 m as a practical walking distance to a transit stop [20–24]. The figures show the relationship between the proposed PUV network and the location of essential facilities in Guimaras. Even when excluding those located in island barangays, there are still numerous essential facilities on the main island that remain underserved by public transport. This lack of direct service is especially evident in the northern and western portions of the island, where clusters of schools, health centers, and commercial areas fall outside the immediate reach of the proposed routes.

Informal modes of transport (i.e., habal-habal) are widely available throughout the island. However, relying on these modes raises several concerns. First, fare rates for these modes are often unregulated and considerably more expensive compared to standard public transport, placing a financial burden on daily commuters, especially those from lower-income households. Second, the safety of these modes is a critical issue. Passengers are often exposed to unsafe road practices, a lack of proper safety equipment, and overloading. These conditions increase the risk of road accidents. These services are also not ideal for transporting the elderly, children, or those with mobility challenges because of their limited space and poor stability.

Given these limitations, it is important that underserved areas be included in the formal public transport network. Even if service is limited to peak travel periods, the availability of more affordable and regulated public transport options would provide residents with a safer and more accessible alternative. This would allow people to plan their trips around scheduled services, improving mobility while reducing dependence on high-cost, informal modes.

To analyze this more thoroughly, the straight-line distances between essential facilities and the nearest PUV routes were measured, and the distribution is presented in Figure 8, which is derived from the facility maps previously shown in Figures 7a and 7b. These earlier figures served as the spatial basis for identifying the proximity of each facility to the proposed transit network. For elementary schools, the distance data reveal a wide dispersion, with numerous outliers extending far beyond the typical range. This indicates that while many schools are situated near transport routes, a notable number are in more remote areas. This spatial pattern reflects a common practice in rural provinces where elementary schools are established near clusters of residential populations to enable young students to walk to school without requiring vehicular transport. However, this setup does not accommodate the mobility needs of

teachers, who are often assigned to distant postings far from their homes. Equipado and Asis-Gilbas [43] highlight that, in such cases, teachers frequently purchase motorcycles, rent tricycles, or ride habal-habal to reach their schools. These transportation options, while necessary, are financially burdensome and physically risky, especially during the rainy season when roads become slippery and hazardous. Thus, there is a need for better PUV access to these locations.

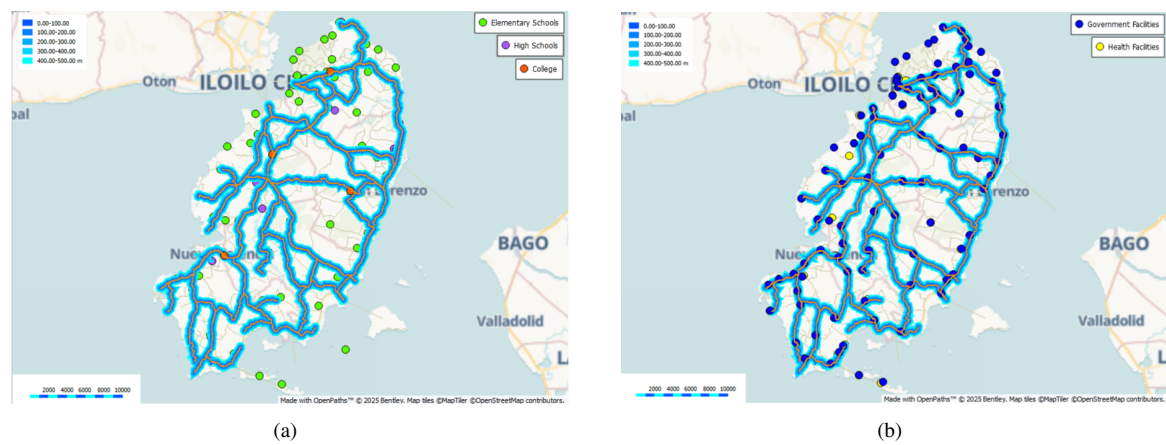


Figure 7. (a) Location of schools with respect to public utility vehicle (PUV) routes; (b) location of government and health facilities with respect to PUV routes

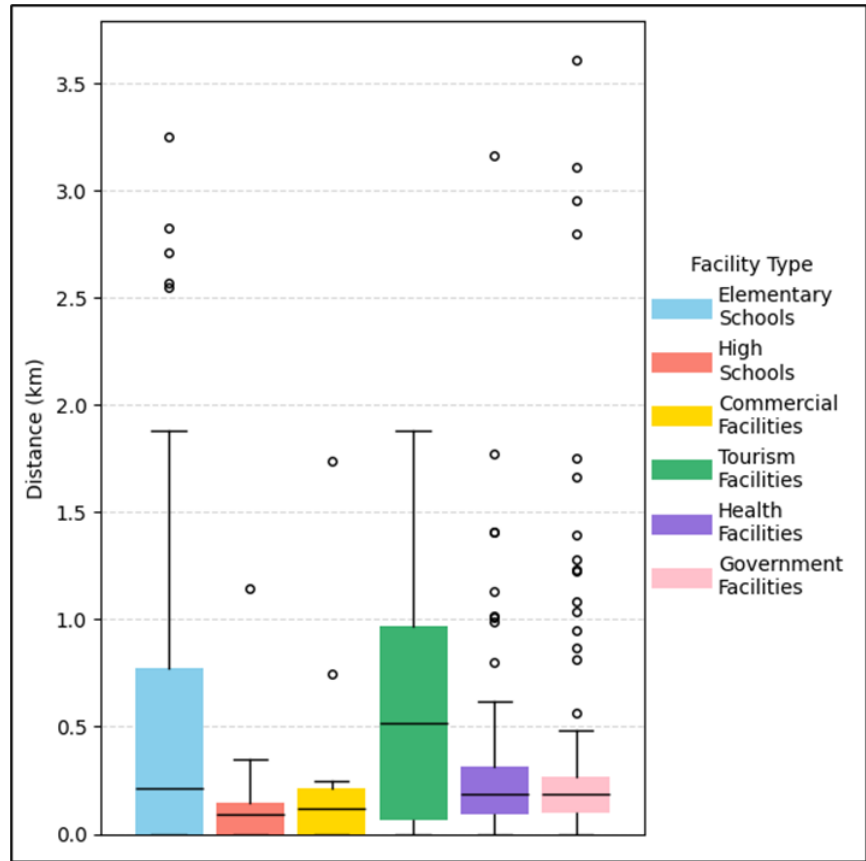


Figure 8. Comparison of distances to the nearest public utility vehicle (PUV) route

In contrast, high schools demonstrate a much more compact distribution of distances, with values tightly clustered near zero. This indicates that most high schools are near PUV routes, enhancing their accessibility for both students and teachers. The narrow interquartile range and minimal presence of outliers suggest a relatively equitable spatial arrangement, likely due to the smaller number of high schools compared to elementary schools and the need to

centralize secondary education facilities for broader catchment areas. In practice, this means students may have to travel longer distances compared to their elementary counterparts, but their access to public transport infrastructure is generally more direct and reliable. In this study, commercial establishments are defined primarily as malls and public markets for simplicity. The measured distances from these facilities to the nearest PUV routes are generally low, indicating good overall accessibility. This aligns with the common practice of situating commercial areas along major roads to maximize visibility, foot traffic, and transport connectivity. However, the presence of a few substantial outliers suggests that some markets or commercial nodes are in more isolated areas, potentially limiting access for both vendors and consumers. These cases highlight the need for better transport linkages to peripheral economic hubs.

The accessibility of tourism facilities varies widely, with distance values ranging from near-zero to nearly two kilometers. This wide dispersion highlights the nature of tourism infrastructure, which is often situated in coastal or mountainous areas that are not necessarily close to main transport corridors. While such locations enhance the appeal of tourist destinations, they also create access barriers, especially for visitors without private vehicles. For local economies that depend on tourism, poor integration with PUV routes can limit visitor volumes and undermine the potential for inclusive growth. Additionally, the reliance on *habal-habal* or special trips for last-mile access raises issues of affordability and safety, particularly for local workers employed in these tourism zones.

The dataset for health facilities reveals a moderate spread in distances, with most values clustered at the lower end and only a few outliers. This suggests that many health centers are relatively well-positioned near PUV routes. The small number of distant outliers may represent rural clinics or health posts located in more remote barangays. Like the spatial logic behind school placements, health facilities in rural provinces are often established near clusters of residential populations to ensure that basic healthcare is locally available. However, this also creates access challenges—not for patients alone, but for healthcare providers as well. Doctors and medical staff are frequently required to travel long distances to reach these remote postings, often relying on informal modes of transport, particularly in the absence of direct public transport routes [42]. Moreover, health facilities in these areas are often less equipped to handle specialized or advanced medical needs, prompting patients to travel even farther to reach better-equipped hospitals or diagnostic centers. Nonetheless, the relatively compact distribution supports the idea that primary healthcare is spatially integrated with transport corridors in many areas.

In this study, government facilities are classified to include both barangay halls and municipal halls. These serve not only as administrative centers but also as important proxies for understanding the general accessibility of each barangay. Since barangay halls are typically located at or near the population centers of their respective communities, the measured distances from these facilities to the nearest PUV routes can be interpreted as indicative of how accessible a barangay is in general. The data show a widespread in distances, with several barangays located far from PUV routes, suggesting significant variation in how connected different areas are to the broader transport network. While some barangays are located near main corridors where PUV routes are readily accessible, others are more isolated due to the absence of nearby routes, making it difficult for residents to access services or travel to other parts of the province. This uneven accessibility highlights the need for more inclusive transport planning that ensures all barangays, especially those on the periphery, are better integrated into the public transport system.

Taken together, the accessibility patterns identified across all facility types point to a structural condition of constrained access, wherein the absence of formal public transport along key corridors inconveniences residents and systematically narrows the range of life opportunities available to them depending on where they live. As Farrington [38] argues, accessibility is not solely an attribute of place but an experience of people relative to their circumstances. This means that the same physical distance to a facility carries vastly different consequences depending on whether the traveler is a low-income daily commuter, a student, an elderly patient, or a teacher on deployment to a remote posting. In Guimaras, this people-centered dimension of accessibility is particularly evident in how the burden of service gaps falls disproportionately on those least equipped to absorb it: lower-income households who cannot afford the unregulated fares of informal modes, essential service providers who must personally shoulder the cost and physical risk of reaching remote assignments, and vulnerable groups such as the elderly and children for whom *habal-habal* travel poses genuine safety concerns. Furthermore, the gaps identified are not simply inefficiencies to be corrected through route adjustments but represent a failure to meet a socially expected standard of access that a public transport system ought to provide. However, it must be acknowledged that greater accessibility is not in itself a sufficient condition for an equitable transport solution, as the economic costs and broader implications of extending service must also be carefully weighed [44]. This tension is particularly pronounced as the barangays most underserved by formal routes are also those that present the greatest operational challenges for service providers (i.e., longer travel distances, lower passenger demand, and road conditions that raise safety risks for operators). Addressing these areas through formal route extension, therefore, requires genuine consultation with and consideration of operator viability, recognizing that sustainable service provision in peripheral areas depends on solutions that are workable for those who must deliver them [45]. The framing of accessibility policies should take place within a common framework that directly involves all relevant stakeholders, ensuring that the needs of

communities and the constraints of service providers are addressed together rather than in isolation [38].

4.3 Route Rationalization

Another key feature of the PUVMP is industry consolidation. Under this requirement, transport operators must form a cooperative, consortium, or corporation before being granted a franchise. This shift is mandated because the current franchising guidelines follow a “one route, one franchise” model, awarding the rights to operate a particular route to a single juridical entity rather than to multiple individual operators.

While this approach is intended to streamline operations and promote more efficient route management, it may lead to unintended challenges. One major issue is fleet coordination in areas with overlapping routes. As shown earlier in Figure 1, the approved inter-municipality routes in Guimaras reveal several cases of significant route overlap. To better understand the magnitude of overlapping routes, Figure 9 visualizes the overlap ratio of all routes within Guimaras Island. A substantial portion of routes exhibit high overlap ratios, with an average of 0.648. The average overlap ratio of 0.648 observed across the Guimaras network is notably high when considered against values reported in comparable transport studies. Literature from other Asian contexts suggests overlap ratios typically ranging from 0.5 to 0.7 [46, 47], though it is important to note that these figures are predominantly drawn from urban settings where a degree of route overlap is more operationally acceptable. In urban areas, the presence of diverse destinations, shorter travel distances, and high passenger density makes it feasible to introduce numerous shorter routes that inevitably share segments of the network [48]. In contrast, rural networks such as that of Guimaras are characterized by fewer destinations and longer travel distances. In these settings, passengers predominantly travel long distances between regions rather than making short intra-regional trips, making route overlap a greater inefficiency relative to the limited demand base. Thus, fewer and longer routes are generally more appropriate in this context, and route overlap carries a greater efficiency cost. This is further underscored by the standards set in study [49], which specifies that new routes shall be designed such that overlaps with existing routes do not exceed 25% of the length of any affected route, and that previously authorized routes may be restructured through merging, shortening, or splitting where necessary. This suggests a level of redundancy within the network wherein multiple routes are relying on the same road segments rather than expanding coverage to underserved areas. Several routes demonstrate near or complete overlaps, meaning their entire alignment coincides with other existing routes. This is particularly evident along key corridors where multiple routes are layered over the same road segments. For example, along the Nueva Valencia–Jordan corridor, several routes follow the same alignment. This occurs even along secondary routes, such as the Guimaras Circumferential Road, where at least three routes overlap on the same segments. Such overlap suggests that route duplication is present across different levels of the road hierarchy, reinforcing that the network is heavily corridor-focused rather than spatially distributed. In turn, this creates competition among drivers, increases the risk of accidents, dilutes passenger demand per route, complicates vehicle scheduling, and leads to underutilization during off-peak periods, further straining financial viability. Consequently, the scenario undermines the intent of consolidation by reintroducing the inefficiencies it seeks to eliminate. On the other end of the spectrum, some routes show relatively low overlap ratios, indicating segments that are more unique. These routes tend to serve the more rural and less connected areas of the island and are critical in improving spatial coverage across the island. According to Arcaño et al. [13], eradicating route overlap is one of the main goals of the LPTRP; however, their analysis of Northern Iloilo also identified numerous overlapping route segments. This emphasizes that route redundancy is not a unique problem in the Guimaras LPTRP but instead is a systemic issue across different cities and provinces. This links back to the hesitation of drivers, as identified by Mateo-Babiano et al. [16], to consolidate routes for fear of reduced financial gains and, at worst, losing their livelihoods. However, this hesitation can be mitigated if drivers are paid a consistent salary or if profit sharing is implemented across a cooperative. A fixed salary and profit-sharing system each present advantages and disadvantages in shaping the attitudes and behavior of drivers in developing countries. Nonetheless, a more structured salary scheme for drivers is generally associated with improved public transport operations [50].

With respect to route structure, one planning approach to addressing route overlap and inefficient fleet deployment is to redesign the configuration of the routes themselves. Specifically, reducing the overall length of individual routes and instead establishing designated stopover or transfer points at key locations across the island. This framework would conceptually allow for the creation of shorter, more manageable routes that connect to specific interchange hubs, rather than long, overlapping lines that span multiple municipalities. Figure 10 presents the spatial distribution of passenger boarding and alighting activity across Guimaras. The figure employs proportional circles scaled to passenger volume, where initial boardings are represented in red, transfer boardings in green, transfer alightings in blue, and final alightings in orange. Transfer points were identified through spatial aggregation and visualization within the software environment, using origin-destination matrix outputs to locate nodes exhibiting concurrent concentrations of both transfer boardings and alightings. Locations where transfer activity was consistently elevated relative to surrounding nodes, and where the clustering of passenger movement patterns was most pronounced, were designated as major interchange points. Since no fixed numerical threshold was imposed, classification instead

relied on relative transfer intensity observed across the network. It is acknowledged that the absence of an explicit quantitative criterion is a limitation, and future studies may benefit from defining formal thresholds to improve the reproducibility and transparency of transfer point identification. As seen in the figure, these concentrations are most pronounced in the central portion of the island, particularly around the municipalities of Jordan and the provincial center, indicating a high volume of passengers changing routes at these nodes. This pattern is consistent with the nature of rural transport, which more commonly serves inter-city connections fulfilling regional mobility needs rather than localized intra-city movement [51]. As such, the high-volume transfer nodes identified in this study reflect points where movement between different parts of the island converges, reinforcing their suitability as locations for formal transfer hubs. This is further supported by Frank et al. [36], whose analysis of multimodal mobility hub placement found that hub density tends to be highest along main roads and key transit corridors, as the existing public transport network naturally shapes where interchange activity is most concentrated. The hotspots identified in Guimaras, corresponding to major junctions along the Nueva Valencia–Jordan corridor, follow a similar logic, reflecting the natural convergence points of travel activity across the island. Beyond the central core, smaller transfer clusters are also observed at peripheral nodes such as Buenavista in the north, San Lorenzo in the east, and Nueva Valencia in the south. These secondary transfer points correspond to major route junctions along the circumferential road network, where passengers transition between routes serving different parts of the island. Formalizing these nodes as interchange hubs has the potential to meaningfully improve access to essential destinations. The introduction of multimodal mobility hubs notably improved accessibility to key points of interest such as health facilities, markets, and workplaces, with accessibility gains observed across many essential destination categories and an associated increase in public transport uptake [36]. Trips to essential destinations benefited the most from hub integration, while non-essential destinations, such as entertainment facilities, remained more difficult to reach within reasonable travel times. In the Guimaras context, this suggests that strategically placed transfer hubs would have the most significant impact on access to the facility types identified in this study as most underserved, while also providing the structural basis for a more deliberate reorganization of how jeepney services are deployed across the island.

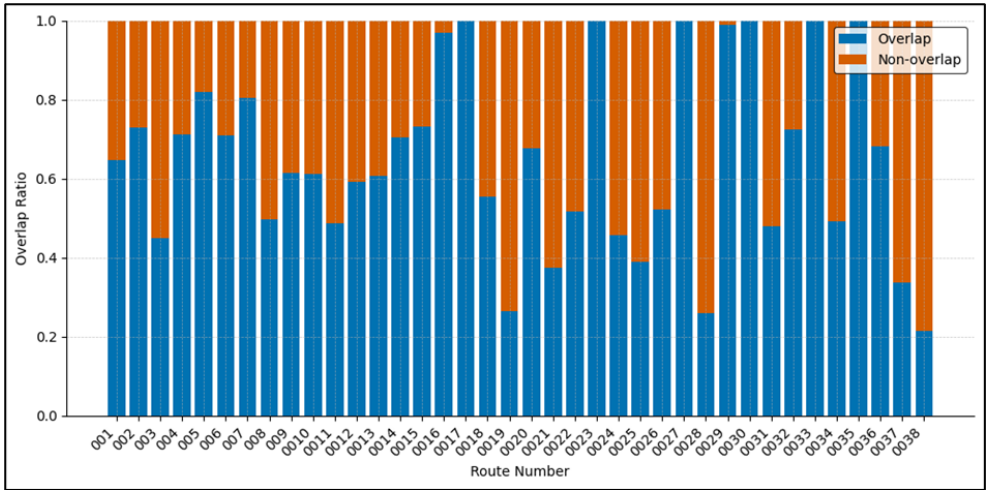


Figure 9. Overlap ratio of all routes

Understanding this reorganization requires situating the jeepney within the broader literature on paratransit. Paratransit refers broadly to passenger transport services that fall outside conventional formal transit and private transport, encompassing shared for-hire services with varying combinations of fixed or flexible routes and schedules [52]. In the Philippines, the jeepney represents the most prevalent form of paratransit and is formally classified as a public utility vehicle. Functionally analogous to a minibus, jeepneys operate along fixed routes but retain the informal characteristic of allowing passengers to board or alight at virtually any point. While convenient, this practice contributes to traffic inefficiencies and inconsistent service quality [53]. Critically, the route structure of jeepney services in Guimaras mirrors a pattern documented extensively in the Global South, where paratransit networks develop through the historical accumulation of individual independent routes, each implemented organically as new areas seek transport access rather than through deliberate network planning [54, 55]. This logic explains the corridor dependency and route overlap observed in the Guimaras LPTRP. Similar to how paratransit networks in Latin American cities converged along primary radial corridors as independently established routes layered over one another toward central destinations [56], jeepney routes in Guimaras have likewise concentrated along the Nueva Valencia–Jordan corridor, reproducing the inefficiencies of demand-driven route accumulation at a provincial scale.

Addressing this requires moving beyond the current direct-service model toward a more structured trunk-and-

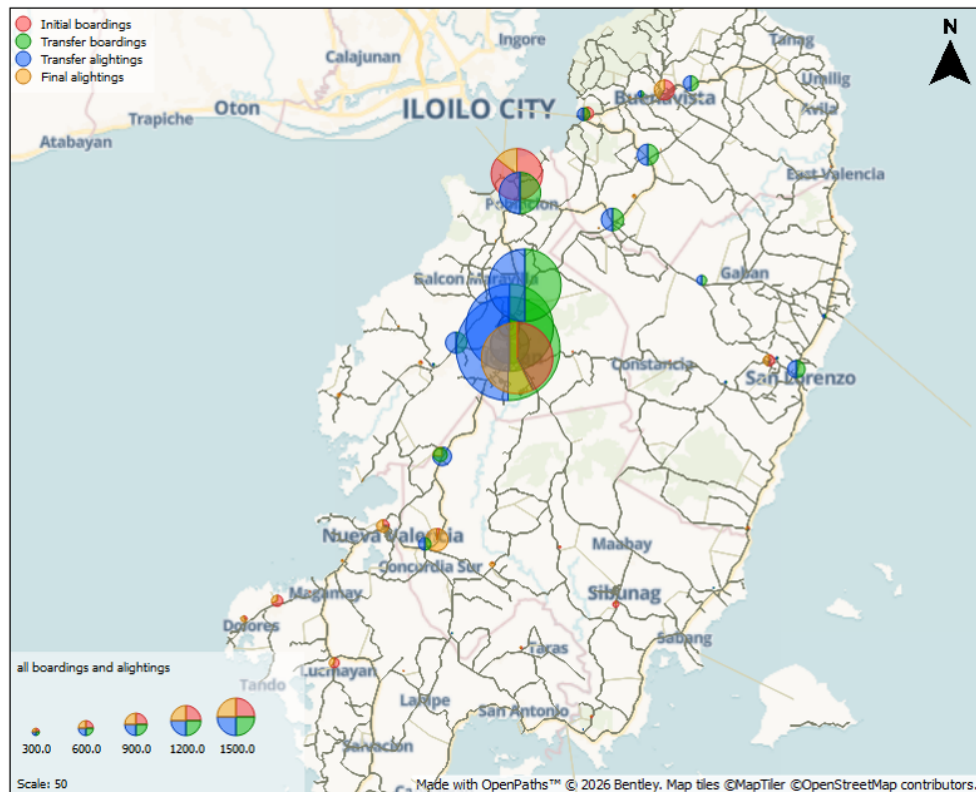


Figure 10. Major boarding, alighting and transfer points in Guimaras

feeder arrangement. Salazar Ferro and Behrens [56] document how the transition from direct paratransit services to feeder-trunk-distributor models can improve network efficiency and reduce route redundancy when implemented with adequate attention to peripheral coverage. Crucially, they argue that paratransit services cannot simply be removed from these systems since doing so would reduce mobility for low-income populations and generate significant operator resistance. Instead, it must be deliberately integrated as complementary feeder services within a structured network [56, 57]. Evidence from Cebu City similarly demonstrates that jeepneys can be effectively repositioned from primary carriers to feeder services supporting a larger transit network, though this transition requires careful management of operator impacts [58]. In the Guimaras context, the proposed transfer hubs should therefore be understood as both passenger convenience infrastructure and organizational anchors of a restructured network where shorter feeder routes from peripheral barangays connect to mainline services along primary corridors. This hub-and-spoke arrangement would replace the current overlapping direct-service model with a more spatially distributed system, provided that appropriate integration measures covering fare systems, intermodal facilities, and service scheduling are in place to ensure that the transition improves rather than disrupts access for communities currently dependent on informal transport [59, 60].

Establishing these transfer points as formal interchange hubs can offer multiple benefits. First, they help minimize unnecessary route overlaps by allowing different routes to terminate or connect at fixed boundaries. This could help ensure that each PUV operator serves a defined geographic segment, rather than overlapping with others for most of the journey. Second, it enables more efficient fleet scheduling, since routes become shorter and easier to manage, especially for cooperatives or corporations operating under the “one route, one franchise” system. Third, it gives passengers more flexibility. Instead of waiting for a single vehicle to complete a long trip across the province, they can transfer between routes at designated points, ideally supported by aligned service schedules and waiting areas. In the long term, implementing this form of nodal or hub-and-spoke transport system may help alleviate congestion at terminals, reduce operational costs for operators, and improve service reliability.

Moreover, a major component of the LPTRP lies in fleet management, specifically how vehicles are deployed, how routes are organized, and how service coverage is maintained throughout the day. In developing a public transport system suitable for Guimaras, it is important to consider local realities, such as dispersed rural settlements, uneven demand across routes, and the limitations of current vehicle types. To explore how different organizational arrangements might address these challenges, three conceptual planning frameworks are discussed below. These are not intended as operationally evaluated scenarios but rather as illustrative frameworks to guide transport planning discussions, drawing from the network analysis and accessibility findings presented in this study.

In one potential scenario, transport services operate under the existing “one route, one franchise” policy implemented by LTFRB. Applying this to Guimaras would require issuing 41 separate franchises to cover the entire island, with each route independently operated. While this approach ensures that each service area has a dedicated operator, it also leads to significant overlaps along the national roads, especially where routes from different municipalities converge. Most of these overlaps occur when vehicles that originate from interior barangays reach the main highway, resulting in multiple units operating along the same corridor. This concentration would likely raise operational challenges, including route congestion and competition for passengers. While assigning staggered stops for overlapping routes can minimize the bunching of vehicles at pick-up and drop-off points, managing dispatch and headway intervals remains complex, particularly in shared segments. In this setup, dispatching may still be controlled at the terminal level, given the unique origins of most routes. However, off-peak service may remain inconsistent, as drivers often wait to fill their vehicles to a breakeven capacity before departing. This operational behavior, while responsive to real-time demand, undermines schedule reliability and limits the accessibility of public transport for passengers traveling outside peak hours.

An alternative planning framework involves grouping routes that share major corridors, particularly along the national roads, into consolidated franchise clusters. Rather than treating each route as an isolated operation, this approach merges complementary services into larger operational clusters (see Figure 11). This aligns with the consolidation objectives embedded in the PUVMP itself, which requires operators to form cooperatives, consortia, or corporations as a prerequisite for franchise issuance, precisely to move away from the fragmented, individually operated model that has historically characterized jeepney services in the Philippines [1]. The highly individualized nature of jeepney operations, where single operators compete along the same routes under a boundary system, leads to on-street competition, uncoordinated services, and route networks that are neither optimized nor integrated. Consolidating routes into grouped franchise clusters directly addresses this fragmentation by enabling common fleet management, coordinated scheduling, and more rational deployment of vehicles across shared corridors. For Guimaras, three main route groups were identified, each formed by consolidating the proposed PUV routes of the Guimaras LGU that share common origin-destination patterns. The first group consists of routes primarily originating from Nueva Valencia and connecting to San Miguel in Jordan, serving the southern municipalities. The second group includes routes from Sibunag and portions of San Lorenzo, which also terminate in Jordan, reinforcing the municipality’s role as a key transport hub. The third group comprises routes from Sibunag and Buenavista, linking to the ports of Buenavista and Jordan, as well as the provincial capital. These terminal points correspond to the main urban centers and port areas, which are identified in the LPTRP as major transfer and interchange locations for both inter- and intra-island travel. This consolidated setup reduces redundancy by streamlining overlapping services, enabling more centralized fleet allocation and scheduling, and supporting a planned and structured transfer system rather than the uncoordinated overlap that currently characterizes the proposed network. High-capacity buses can also be introduced along the main corridors to improve efficiency and reduce operating costs through economies of scale, although some transfers between services may still be required. However, it is worth noting that consolidation in practice has proven difficult to achieve. Sunio et al. [1] document how individual operators in Metro Manila resisted joining cooperatives, citing reluctance to surrender vehicle ownership and distrust in cooperative management, ultimately hampering franchise consolidation and common fleet management. Since operators in Guimaras similarly function under individualized arrangements, these same resistances are likely to emerge, and any consolidation strategy will need to be accompanied by adequate stakeholder engagement, financial incentives, and capacity-building support to be viable [16].

A third conceptual framework considers what a fully integrated arrangement might look like, wherein all transport services are managed under a single provincial-level entity. In this arrangement, route overlaps would cease to be a problem, as all services could be coordinated to complement, rather than compete with, one another. This structure could also support the introduction of key transfer terminals at strategic points across the island, particularly along the Guimaras Circumferential Road, where transit flow patterns indicate frequent transfers. As Salazar Ferro and Behrens [56] note, the viability of such a model depends on achieving operational complementarity between formal and informal services rather than displacing one entirely. Vehicles could potentially be allocated dynamically, shifting to meet fluctuating demand across different times and locations. This structure could also support the introduction of key transfer terminals at strategic points across the island, particularly along the Guimaras Circumferential Road, where transit flow patterns indicate frequent transfers. These terminals would be equipped with multiple bays to accommodate mainline PUVs, as well as feeder modes such as tricycles and habal-habal serving outlying barangays not directly covered by the PUV routes. By centralizing control of the network, this scenario could facilitate better scheduling, improved service coverage, and consistent operations, especially in areas with lower demand. Terminals could also function as coordination hubs, improving the passenger experience through reliable transfers and providing sheltered waiting areas, posted schedules, and information systems.

It is important to note that the fleet management scenarios presented in this section are intended as planning guidelines rather than definitive operational management. The primary objective of this study is not to conduct a



Figure 11. Proposed grouping of public utility vehicle (PUV) service routes

detailed financial or operational analysis of each scenario, but rather to offer a critical assessment of the current LPTRP proposal and provide a framework that can serve as a baseline for program implementation at the provincial level. As an exploratory study, the aim is to generate a general discussion that highlights key considerations for applying the PUVMP in an insular context, which can inform future planning efforts. A more rigorous evaluation, including simulation modeling, cost-benefit analysis, and ridership impact assessment, is recommended for future research once sufficient operational data becomes available. In this regard, financial analysis of the scenarios was not pursued, as detailed data on jeepney operating costs, revenue structures, and operator willingness to serve low-demand routes remain limited in the study area. The absence of such data constrains the depth of quantitative evaluation that can be reliably performed at this stage. Furthermore, the transport modeling software employed in this study presents inherent limitations in replicating the grouped and consolidated route scenarios described above. While routes can be segmented and reorganized conceptually, the software ultimately processes each route on an individual basis, meaning that the grouped franchise and provincial consolidation scenarios cannot be fully simulated in a manner that accurately reflects real-world network behavior. Any attempt to model these scenarios within the current framework would yield results that do not meaningfully represent the intended operational structure.

Ultimately, the choice of fleet management strategy will shape the effectiveness and equity of Guimaras' public transport system. Whether through route-level franchising, grouped operations, or centralized provincial control, each model offers trade-offs between operational complexity, cost-efficiency, and user experience. However, aligning route design with realistic demand projections, minimizing redundancies, and investing in transfer infrastructure are key to realizing a system that meets the needs of both operators and the commuting public.

5 Conclusions and Recommendations

PUVMP represents a fundamental shift in how public transport is planned and delivered in the Philippines, moving away from a market-driven, operator-led model toward a structured, LGU-led framework anchored in the LPTRP. While this shift addresses long-standing inefficiencies in route provision, existing studies have largely examined its application at the level of individual LGUs in urban or mainland settings. This study addresses a gap in the literature by evaluating the LPTRP at a provincial island scale, where multiple LGU-level plans are consolidated into a single island-wide framework. It represents one of the first attempts to assess LPTRP implementation at this broader geographic scope, through transport modeling and accessibility analysis. The study examined route coverage, facility access, and network structure across the five municipalities of Guimaras Province. The findings

reveal that the program's current design does not adequately account for the spatial and operational realities of insular settings.

The most critical finding of this study is not simply that service gaps exist, but that the current LPTRP reproduces the same corridor-dependency that characterized the pre-reform system. With an average route overlap ratio of 0.648, most proposed routes concentrate along the same primary corridors, particularly the Nueva Valencia–Jordan corridor, rather than expanding coverage toward underserved areas. This means that the rationalization process, intended to correct inefficiencies, has instead formalized them at the provincial scale. The result is a network that serves the already-accessible central corridors well while systematically excluding peripheral barangays, particularly in the northern and western portions of the island, where clusters of schools, health centers, and government facilities fall outside the 500-meter catchment threshold. This pattern challenges a core assumption of the LPTRP framework, which presumes that route planning grounded in passenger demand will naturally yield equitable coverage. In a dispersed island setting, demand-driven planning alone is insufficient because low-density and remote areas generate lower ridership figures, making them unattractive to operators and therefore chronically underserved under any market-responsive model.

Based on these findings, three policy directions are recommended. First, service provision in underserved areas should be formalized through scheduled low-frequency routes. The accessibility analysis reveals that the burden of service gaps falls disproportionately on essential service providers, particularly teachers and healthcare workers assigned to remote postings. In the Philippine context, these workers are typically deployed on a weekly rather than daily basis, meaning their transport needs, while less frequent than those of daily commuters, are predictable and recurring. Scheduled PUV services aligned with this weekly rhythm can therefore meaningfully address this demand without requiring continuous daily operations. Formalizing these trips within the LPTRP framework would bring them within the regulated network, improving both safety and affordability. While emergency situations introduce variability that fixed schedules cannot fully resolve, the predictability of weekly deployment patterns provides a sufficient operational basis for structured low-frequency service along these corridors.

Second, route rationalization should be reoriented away from corridor consolidation alone and toward spatial redistribution. The current network concentrates services along established corridors, leaving peripheral barangays systematically underserved, resulting in an average overlap ratio of 0.648 across all proposed routes. Establishing formal transfer hubs at identified high-volume nodes, particularly in Jordan, Buenavista, San Lorenzo, and Nueva Valencia, would allow shorter feeder routes to serve outlying areas while connecting passengers to mainline services at interchange points. This hub-and-spoke structure reduces redundancy without sacrificing coverage and provides a more equitable basis for network design. Transfer hubs should be physically designed to accommodate multiple route connections simultaneously, with covered waiting areas, posted schedules, and clear passenger information to facilitate smooth transfers and encourage modal shift away from informal modes. The placement of these hubs should also account for the spatial distribution of essential facilities identified in this study, ensuring that nodes are positioned to maximize access to schools, health centers, markets, and government offices rather than simply reflecting existing traffic concentrations.

Along the main corridors where overlap is highest, route boundaries should be clearly delineated so that each franchise operates a defined segment rather than duplicating the full length of a shared alignment. This would not only reduce competition among operators along the same corridor but also create the operational conditions necessary for reliable headway management, because vehicles that serve shorter, bounded segments are easier to schedule and monitor than those running long cross-provincial alignments. Routes that currently demonstrate low overlap ratios, which tend to serve the more rural and isolated portions of the island, should be retained and protected in any rationalization process, as these represent the network's only formal connection to communities that would otherwise remain entirely dependent on informal transport.

Third, the consolidation model should be revisited at the provincial level. The current one-route, one-franchise approach applied across 41 routes perpetuates the fragmented operational structure that the PUVMP seeks to overcome. A grouped franchise or single provincial entity arrangement would enable more coherent coordination of overlapping services, centralized fleet deployment, and cross-subsidization of low-demand routes through revenues generated along busier corridors. The viability of peripheral services depends on their integration into a broader, managed network rather than their performance as isolated operations. Under a grouped franchise model, cooperatives would have greater flexibility to reallocate vehicles as needed across their assigned corridor, reducing the problem of vehicles waiting at terminals for full loads before departing, which is a key driver of schedule unreliability in the current PUV system.

A provincial-level entity would go further by enabling network-wide scheduling and coordinated headway management across all routes. This would also open the door to more strategic fleet investments, such as the introduction of higher-capacity vehicles along the main Nueva Valencia–Jordan corridor, where demand is demonstrably concentrated, while deploying smaller, more appropriate vehicle types along feeder routes serving lower-density areas. Furthermore, centralized management would facilitate the collection of operational data,

including ridership counts, trip times, and load factors, which are currently unavailable and which severely constrain the ability of regulators to make evidence-based decisions about route design and service frequency. Building this data infrastructure into the consolidation framework from the outset would significantly strengthen the capacity of future LPTRP revisions to respond to actual travel demand rather than relying on outdated or incomplete baseline surveys.

This study is exploratory in nature, and the recommendations presented here are intended as planning guidelines rather than cure-all solutions. In addition, the analysis was constrained by the use of the 2018 household interview survey dataset due to the absence of more recent island-wide origin-destination and travel behavior data. While demand growth adjustments were incorporated to account for temporal changes in travel demand, these projections primarily reflected aggregate traffic growth trends and may not have fully captured post-pandemic behavioral shifts. These may include changes associated with remote work arrangements, such as online education, tourism recovery dynamics, and evolving modal preferences. Further research incorporating updated household travel surveys, operational data from passengers and operators, detailed scenario modeling, and feasibility assessments will therefore be necessary to evaluate the practical implementation and long-term applicability of these recommendations. Nevertheless, the findings contribute a replicable analytical framework for assessing LPTRP proposals in other island-provinces across the Philippines and in comparable insular settings in Southeast Asia, where equitable transportation access remains a challenge.

Author Contributions

Conceptualization, A.F.; methodology, A.F. and A.I.; software, A.F.; validation, A.F., L.M., and A.I.; formal analysis, A.F.; investigation, A.F. and L.M.; resources, A.F.; data curation, A.F. and L.M.; writing—original draft preparation, A.F., L.M., and A.I.; writing—review and editing, A.F., L.M., and A.I.; visualization, A.F. and L.M.; supervision, A.F.; project administration, A.F. and L.M.; funding acquisition, A.F. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

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